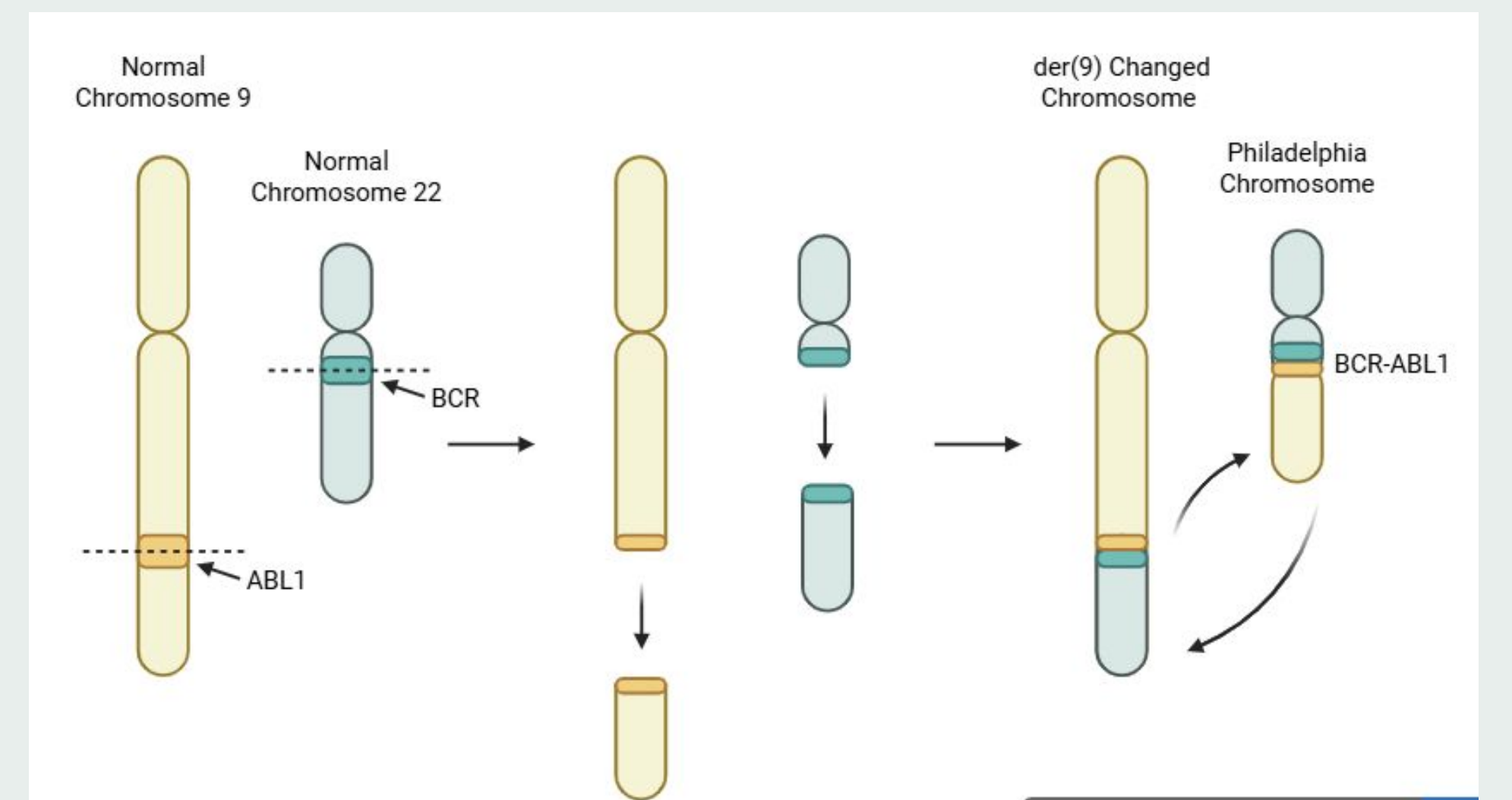


The Combination Therapy of Imatinib and Curcumin To Combat Imatinib-Resistant Chronic Myeloid Leukemia

Kara Wang, Dr. Pallavi Phatale

CML and Imatinib

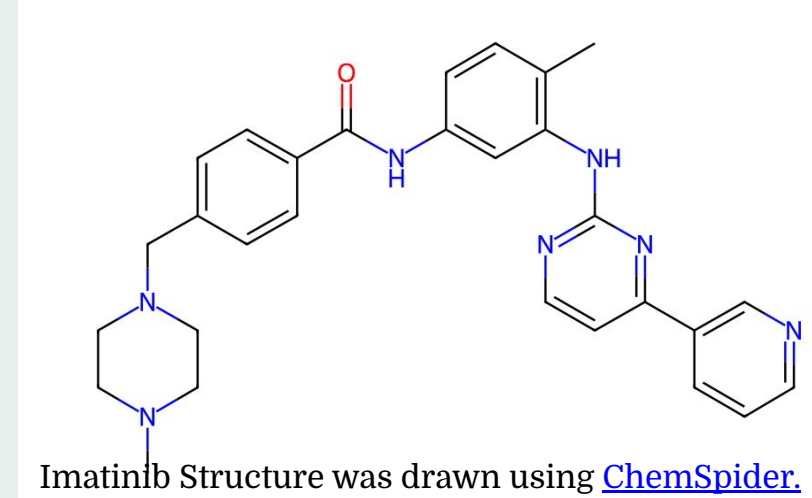
Chronic Myeloid Leukemia is a blood cancer that leads to the overproduction of immature myeloid cells in the bone marrow. Studies show that more than 95% of patients have a chromosomal translocation called Philadelphia chromosome in which chromosome 9 and chromosome 22 are fused together. This type of translocation will create the BCR-ABL1 fusion protein, which is an always active tyrosine kinase that leads to uncontrolled growth of white blood cells.



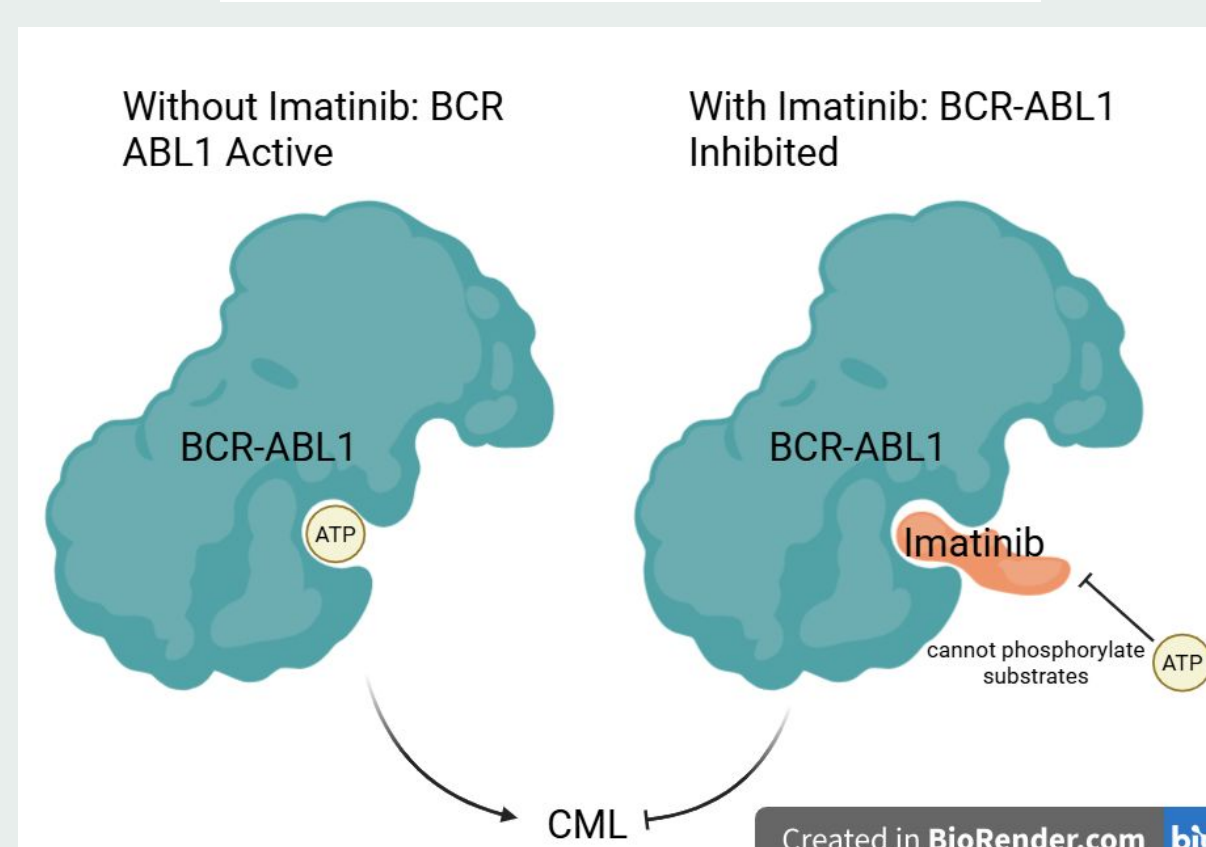
Formation of BCR-ABL1 Fusion Protein (Figure representation created by the author: Kara Wang in BioRender)

Pharmacodynamics of Imatinib

Imatinib is a tyrosine kinase inhibitor (TKI) and is a drug that is widely known to treat Chronic Myeloid Leukemia. It inhibits BCR-ABL1 by blocking its ATP binding site, stopping any further cell proliferation. Although it has been proven to have significantly revolutionized CML treatment, 30% of CML patients have shown signs of resistance towards imatinib. One such resistance mechanism is the T315I point mutation in the BCR-ABL1 ATP binding site, which prevents imatinib from binding.



Imatinib Structure was drawn using ChemSpider.



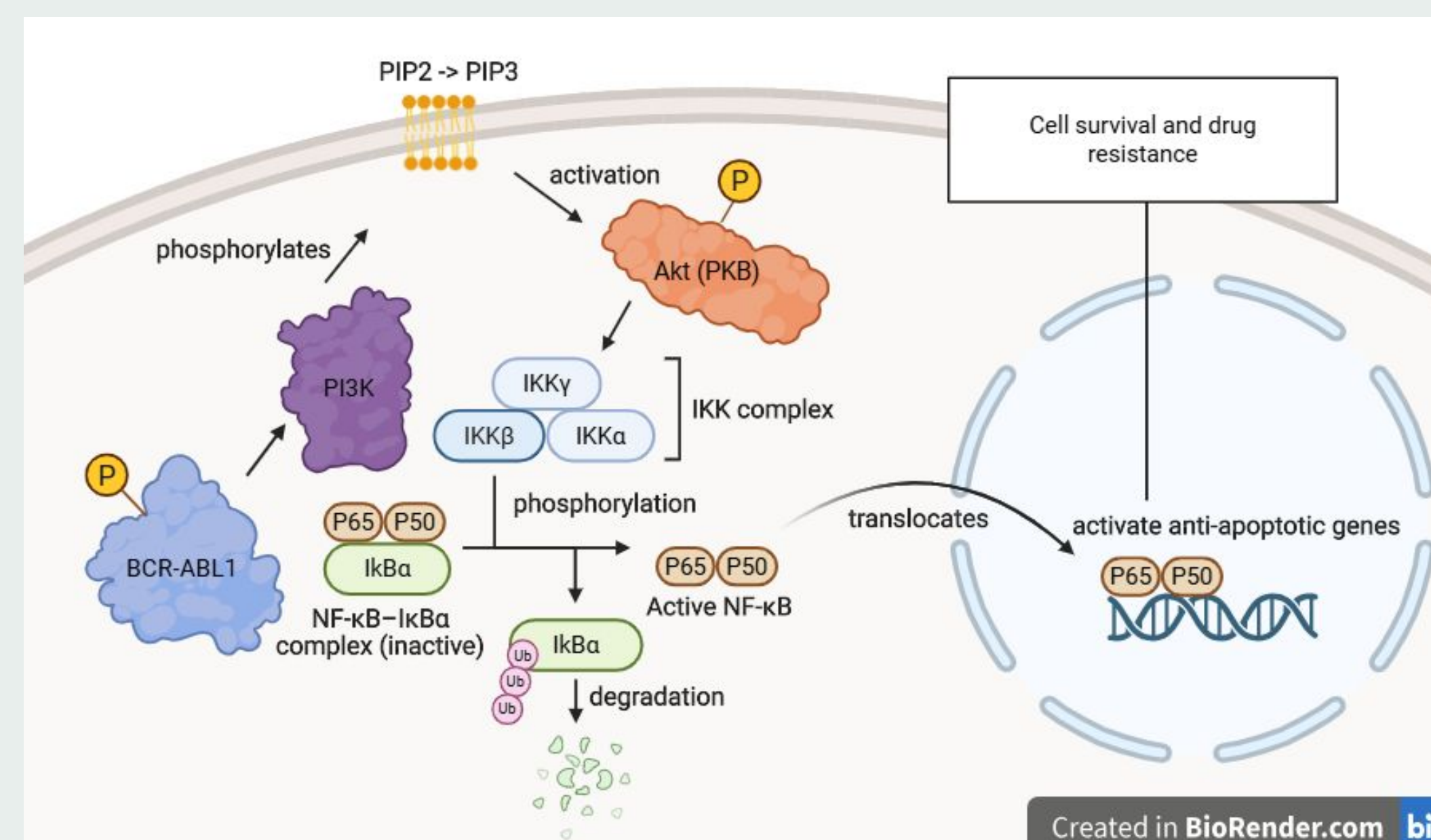
Comparison of BCR-ABL1 With and Without Imatinib (Figure representation created by the author: Kara Wang in BioRender)

CML cells rely on many oncogenic mechanisms to support cancer growth, and one example would be PI3K/Akt/mTOR/NF-κB pathway, which is activated by the BCR-ABL1 fusion protein. It promotes cell proliferation, stops apoptosis, and aids drug resistance. It has also been observed that high nitric oxide (NO) levels have been seen in CML patients. Nitric oxide promotes tumor growth, DNA damage, and resistance to therapy.

Causes of Drug Resistance

Although imatinib inhibits BCR-ABL1 in many patients, resistance occurs in around 30% of patients.

- One resistance example is the T315I point mutation, which alters BCR-ABL1's shape to stop imatinib from binding effectively. This allows cancer cells to continue growing.
- Overexpression of BCR-ABL1 leads to resistance because excess oncoprotein overwhelms the TKIs since there is too much to target.
- BCR-ABL1 also activates several signaling pathways that support cancer cells, including the PI3K/Akt/mTOR/NF-κB pathway. BCR-ABL1 first activates PI3K, a cytoplasmic protein that moves to the cell membrane when stimulated. At the cell membrane, it converts PIP2 to PIP3, activating Akt. Akt then activates mTOR (promotes cell division) and also stimulates the IKK complex, which phosphorylates and degrades IκBα, a protein that normally keeps NF-κB inactive. The now activated NF-κB will enter the nucleus and activate anti-apoptotic genes. Consistent use of this pathway will contribute to drug resistance in CML, which is why researchers are trying to block parts of this pathway.



PI3K/Akt/mTOR/NF-κB Pathway (Figure representation created by the author: Kara Wang in BioRender)

Age Populations That Suffer From CML

15%

of leukemia patients are diagnosed with CML

30%

of CML patients develop imatinib-resistance

The median diagnosis age is around

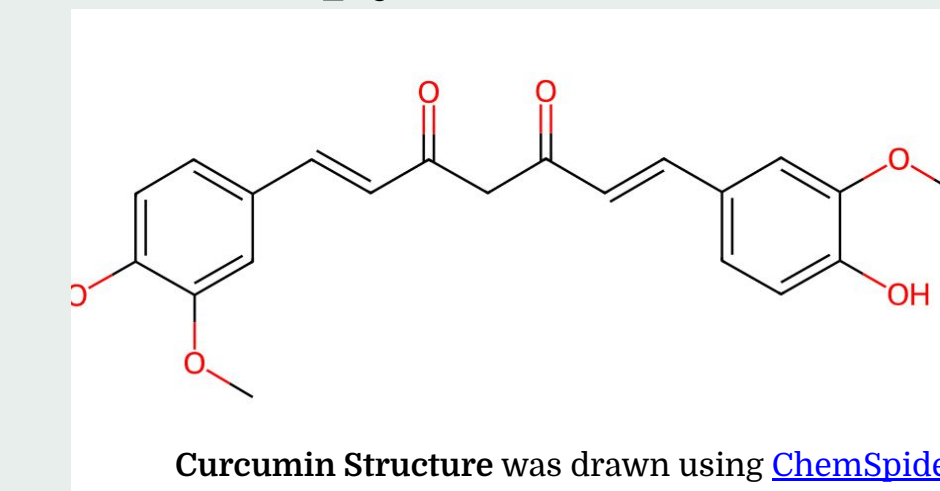
64-66 years

- Most commonly found in adults

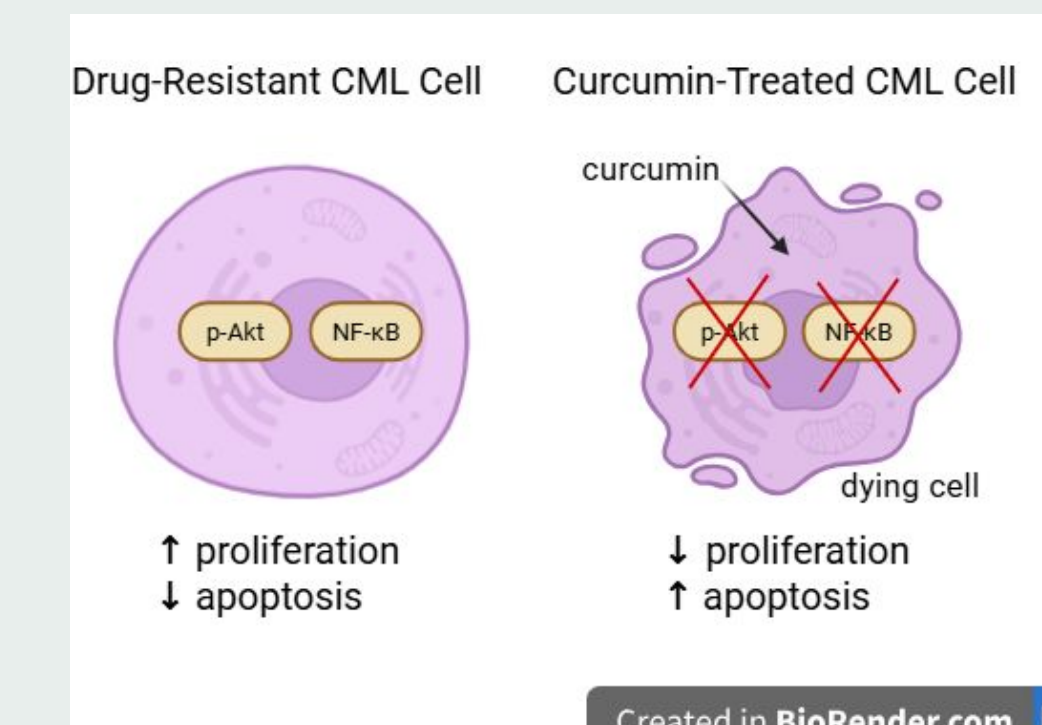
Curcumin

Curcumin is a component of turmeric. It is known for its anti-cancer and anti-inflammatory properties. Because of this and low toxicity, it has been studied for its potential to work as adjuvant therapy with imatinib-resistant CML.

However, curcumin has limitations such as low bioavailability, poor absorption, and rapid metabolism. Because of this, new formulations have been created, including curcumin analogs that are designed to enhance solubility and slow down metabolic processes.



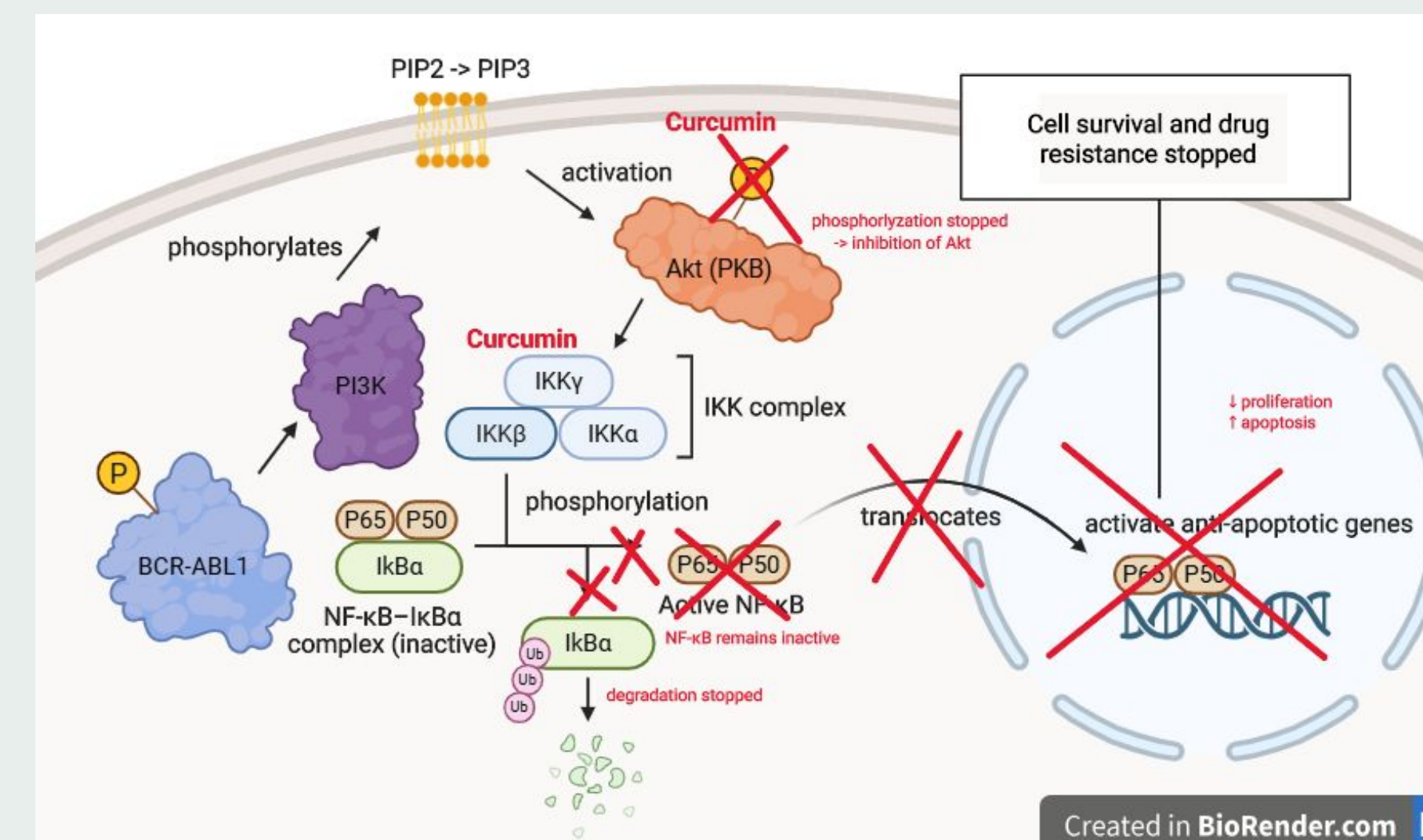
Curcumin Structure was drawn using ChemSpider.



Curcumin Treated and Untreated Comparison (Figure representation created by the author: Kara Wang in BioRender)

Circumin Results On CML

Researchers tested curcumin on both imatinib-sensitive (K562, LAMA84S) and imatinib-resistant (LAMA84R) CML cell lines with different assays. Results suggest that curcumin reduced cancer cell life in a dose and time-dependent manner. Western blot analysis revealed that curcumin decreased the expression levels of p-Akt and p-NF-κB, which promote cell proliferation in the signaling pathway. In resistant cells, Caspase-3 also had an increase in expression, meaning increased apoptosis. Furthermore, through molecular docking assay, curcumin showed a strong binding affinity to NF-κB and Akt, which helps in inhibiting pathways. Curcumin has also shown the ability to suppress NO levels, which are usually high in leukemia.



PI3K/Akt/mTOR/NF-κB Pathway With Curcumin Treatment Effects (Figure representation created by the author: Kara Wang in BioRender)

Imatinib Retention With PK

Since CYP3A4 is an imatinib metaboliser, and both imatinib and curcumin inhibit CYP3A4, curcumin can help slow down the breakdown of imatinib, allowing for imatinib to stay in the bloodstream for longer. This enhances imatinib therapeutic effect without increasing its dose (helps lower toxicity too).

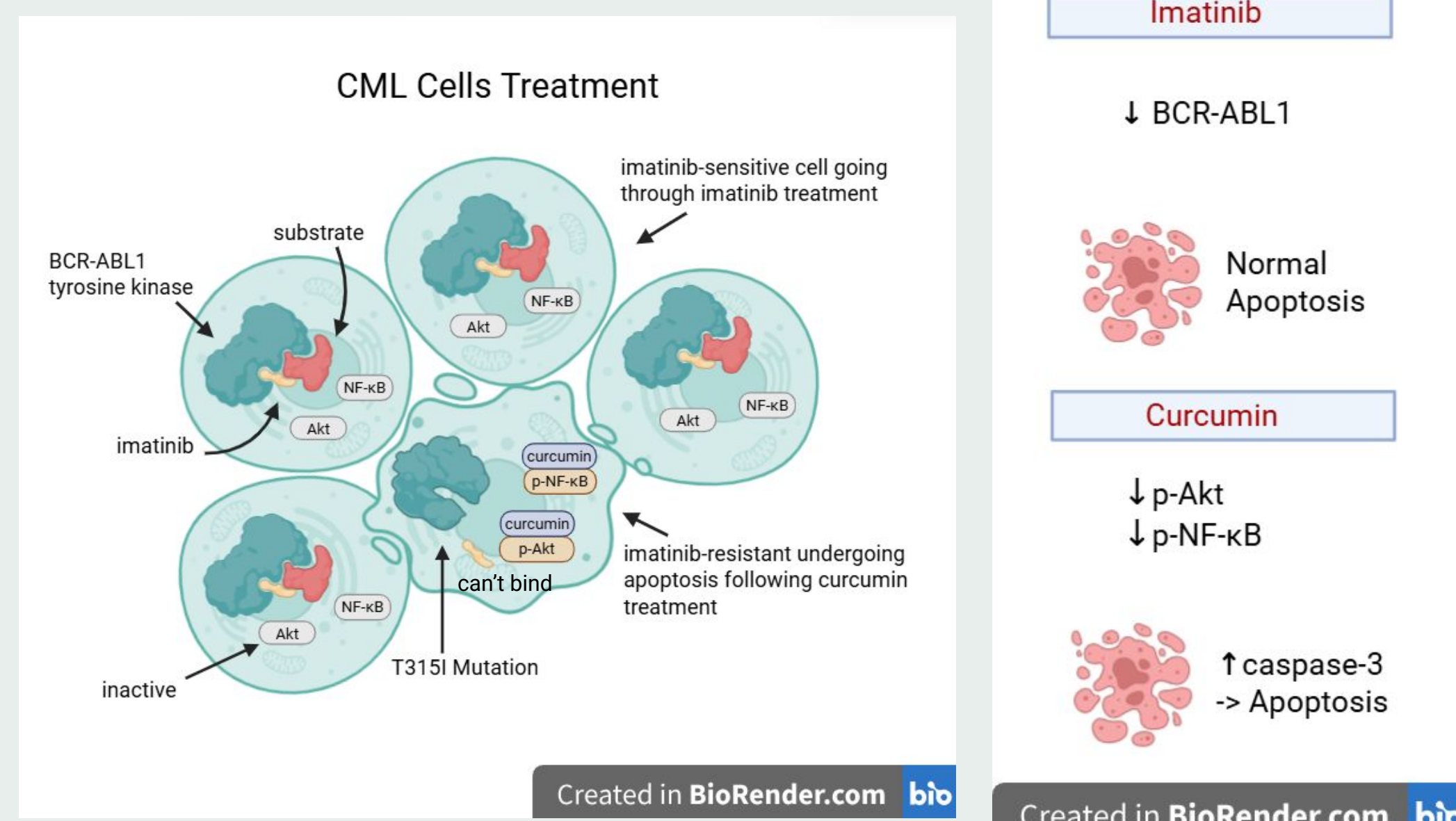
Possible Mechanism to Overcome Drug Resistance

PI3K/Akt/mTOR/NF-κB contributes to resistance, allowing for leukemic cells to continue growing while avoiding apoptosis.

It was shown that curcumin has these abilities:

- Inhibit Akt and NF-κB (reduce cell proliferation)
- Increase caspase-3 expression (for apoptosis)
- Strong binding affinity with NF-κB and Akt (help in inhibiting pro-cancer pathways)
- Lower nitric oxide levels (reduce tumor growth)

Because of these qualities, combining curcumin with imatinib may help overcome the resistance that many CML patients face. Imatinib targets the imatinib-resistant CML cells, while curcumin can target the imatinib-resistant cells. Specifically, for drug-resistant cells, curcumin binds Akt and NF-κB, causing inhibition of the pathway. The higher caspase-3 expression also helps by promoting apoptosis.



Curcumin and Imatinib Working Together in CML Treatment (Figure representation created by the author: Kara Wang in BioRender)

Imatinib and Curcumin Path To Apoptosis (Figure representation created by the author: Kara Wang in BioRender)

Conclusion

Curcumin may help overcome imatinib resistance in Chronic Myeloid Leukemia (CML) by inhibiting cancer survival signaling pathways, reducing nitric oxide levels, and increasing caspase-3 expression, suggesting its potential as combination therapy with imatinib.

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